GroupTrack

V3.0 Implementation Plan — Separated Build Strategy

March 29, 2026 | Branch: feature/convoy-event-ride | Gate: Wednesday Field Test

1. Executive Summary

GroupTrack V3.0 migrates from a fully offline standalone app to a cloud-assisted platform. The migration must be built in two completely isolated phases with a hard gate between them — V2.4 Wednesday field test confirmation — to ensure the proven offline radio engine is never destabilized by backend integration work.

The core principle: build everything that has no device dependency first. Schema, service layer, sign-in screen, and API wiring are all safe to build now. The single connection point — replacing the existing ride creation flow with an API-backed version — is the last thing wired, and only after Wednesday confirms V2.4 is solid.

2. Architecture — What Changes and What Doesn't

2.1 Unchanged in V3.0

COMPONENT	REASON UNCHANGED
Radio write engine	ConvoyProfileBuilder, InstallProfileUseCase, channel write — all V2.4. WorkingConfig source changes (local → downloaded JSON) but merge logic is identical.
Mesh radio operation	All mesh, GPS tracking, HUD, map display, and node tracking is purely offline. No V3.0 change touches any of this.
master.cfg binary	Stays in assets. Not hosted in RDS. Not modified by V3.0.
Offline map tile system	ConvoyTileDownloader, shouldInterceptRequest, convoy:// scheme — unchanged. V3.0 adds automated trigger, not new downloader.
Developer settings gate	Long press header → hardcoded password. Completely independent of Google Sign-In. Do not entangle these two auth systems.
Local JSON cache structure	convoy_events/{UUID}.json format unchanged. Downloaded rides save here identically to .convoy email imports.

2.2 What V3.0 Adds

NEW COMPONENT	DESCRIPTION
RDS Schema	4 tables: users, rides, invites, enrollments. Created on EC2/RDS before any Android work.
PHP Service Layer	convoy_api.php already live. Endpoints: POST /users/register, POST /rides, POST /rides/{id}/invite, GET /ride/{token}, POST /enroll.
ConvoySignInScreen.kt	New screen. Google One Tap auth. POSTs to /users/register on success. Caches user_id in SharedPreferences. Not yet wired into navigation.
ConvoyApiClient.kt	New file. OkHttp wrapper for all API calls. No business logic — pure

	HTTP. Isolated, testable, not wired to UI until Phase 2.
API_BASE_URL in ConvoyConfig	const val API_BASE_URL = "http://34.224.89.217/" — one line, no functional impact.
Invite deep link handler	AndroidManifest intent-filter for grouptrack.org/invite/. Extracts token, calls GET /ride/{token}, saves JSON to convoy_events/.

2.3 The Single Connection Point

There is exactly one place where V3.0 wires into existing V2.4 code: the ride creation flow. In V2.4, Create a Ride saves a local JSON file. In V3.0, it POSTs to /rides and saves the response to local cache. Everything downstream — WorkingConfig, radio write, RIDE tab display — is identical because both versions produce the same local JSON.

> ⚠️ **This connection point is ONLY wired after Wednesday test confirms V2.4 is stable. All other V3.0 work is built in isolation before that gate.**

3. Phase 1 — Isolated Build (Safe Now, No Device Risk)

Everything in Phase 1 can be built immediately. None of these tasks touch existing V2.4 screens, ViewModel state, or the radio engine. If Phase 1 code has a bug, it has zero impact on the working V2.4 build.

3.1 Backend — RDS Schema

Task B-01 — Create RDS Schema

ITEM	DETAIL
What	Run create_schema.sql against convoy_tracker database on RDS. Creates 4 tables: users, rides, invites, enrollments with all foreign keys.
Where	EC2 SSH session → mysql -h convoy-tracker-db.cudtjxrtdbql.us-east-1.rds.amazonaws.com -u convoy_admin -p convoy_tracker < create_schema.sql
Verify	SHOW TABLES; — expect 4 rows. DESCRIBE users; DESCRIBE rides; — confirm columns match spec.
Risk	**NONE — pure database, no Android code**
Est.	30 minutes

Task B-02 — Verify PHP API Endpoints

ITEM	DETAIL
What	Confirm all 5 endpoints return expected responses after schema is created. Test with curl from Git Bash.
Test	curl -X POST http://34.224.89.217/users/register -H "Content-Type: application/json" -d '{"google_id":"test123","email":"test@test.com","first_name":"Fred","last_name":"Test"}'
Expect	JSON response with user_id UUID. Verify row appears in RDS: SELECT * FROM users;
Risk	**NONE — server-side only, no app impact**
Est.	20 minutes

3.2 Android — New Files Only (Zero Touch to Existing Code)

Task A-01 — Add API_BASE_URL to ConvoyConfig.kt

ITEM	DETAIL
What	Add const val API_BASE_URL = "http://34.224.89.217/" to ConvoyConfig.kt.
File	app/src/main/java/com/geeksville/mesh/convoy/ConvoyConfig.kt
Impact	One line addition to an existing constants file. No behaviour change. Nothing calls it yet.
Risk	**NONE**

Est.	5 minutes

Task A-02 — Create ConvoyApiClient.kt (New File)

ITEM	DETAIL
What	New file. OkHttp-based API client. Suspend functions for each endpoint. No UI dependency. No ViewModel dependency.
File	app/src/main/java/com/geeksville/mesh/convoy/ConvoyApiClient.kt — NEW
Functions	registerUser(googleIdToken), createRide(rideData), createInvite(rideId), downloadRide(token), enrollRider(rideId). Each returns Result<T>.
Impact	New file only. Not imported by any existing file. Zero risk to V2.4 operation.
Risk	**NONE — new file, not referenced by anything yet**
Est.	1.5 hours

Task A-03 — Create ConvoySignInScreen.kt (New File)

ITEM	DETAIL
What	New Compose screen. Google One Tap integration. On success: calls ConvoyApiClient.registerUser(), caches user_id + google_id + name in SharedPreferences.
File	app/src/main/java/com/geeksville/mesh/convoy/ConvoySignInScreen.kt — NEW
Dependency	Add to build.gradle: com.google.android.gms:play-services-auth:21.x.x — already listed in V3 Architecture Plan.
Navigation	NOT yet wired into nav graph. Screen exists but is not reachable from the app. Tested directly via unit test or temporary debug entry point only.
Impact	New file + one build.gradle line. No existing screen or ViewModel touched.
Risk	**NONE — unreachable screen cannot break existing flow**
Est.	2.5 hours

Task A-04 — Add Invite Deep Link Handler

ITEM	DETAIL
What	Add intent-filter to AndroidManifest for grouptrack.org/invite/ deep links. Handler extracts token, calls GET /ride/{token}, saves JSON to convoy_events/ using existing import handler.
Files	AndroidManifest.xml (intent-filter only) + new ConvoyInviteHandler.kt for token extraction and ride download logic.
Safe boundary	Downloaded ride JSON is saved to convoy_events/ identically to a .convoy email import. The existing import handler processes it. No new code path for ride loading.
Risk	**LOW — AndroidManifest change requires careful testing. Deep link must not intercept unrelated URLs.**
Est.	1.5 hours

Task A-05 — Lead Lock UI Buttons

ITEM	DETAIL
What	Add SET AS LEAD button to NODE panel. Add I AM LEAD and RECALC LEAD buttons to MY CART panel. Logic already committed — buttons are UI only.
File	ConvoyScreen.kt only. Calls viewModel.setExplicitLead() and viewModel.recalcLead() — both already exist.
Risk	**LOW — touches ConvoyScreen.kt which is the main screen. Must build and test before Wednesday.**
Est.	1.5 hours

4. Wednesday Gate — V2.4 Field Test

This is a hard gate. Phase 2 wiring does not begin until all Wednesday tests pass. The gate exists to protect the proven offline radio engine from being destabilized by backend integration.

#	ID	Test	Gate Status
1	**FT-01**	Lead lock — drive 1/4 mile after RECORD, verify lead locks, single clean track line on switchbacks.	**PENDING**
2	**FT-01B**	SET AS LEAD, I AM LEAD, RECALC LEAD buttons — verify all three overrides work correctly.	**PENDING**
3	**FT-04**	Offline maps render cleanly at z18 across downloaded SAT area. No grey patches.	**PENDING**
4	**FT-05**	Maps render full screen at all zoom levels. No half-screen or spotty tile display.	**PENDING**
5	**FT-06**	Auto-pan — zoom and pan do not snap back. Search result stays centered. GROUP/MY CART re-centers correctly.	**PENDING**
6	**FT-02**	Channel utilization stable at 23-25% with 5 nodes. Lead at 3s interval, others at 5s.	**VALIDATED — no retest needed**
7	**FT-03**	Bluetooth stability with Unrestricted battery setting. No timeout during ride.	**DOCUMENTED — workaround in place**

⚠️ If FT-04 or FT-05 fail, offline maps must be resolved before Phase 2 begins. Do not start RDS wiring on a build with broken offline maps.

5. Phase 2 — Integration Wiring (Post-Wednesday Only)

Phase 2 wires the isolated Phase 1 components into the existing V2.4 app. Each task has a clearly defined connection point — the exact line of code or screen where the new component touches existing code for the first time.

5.1 Wire Sign-In into App Launch

ITEM	DETAIL
Connection point	MainActivity.kt or ConvoyScreen.kt LaunchedEffect on startup — check SharedPreferences for cached user_id. If missing, navigate to ConvoySignInScreen.
Key rule	If sign-in fails or is unavailable, app proceeds to main map anyway. Organizer features show a 'Sign in required' banner. Radio operation is never blocked.
Files touched	MainActivity.kt or top-level nav graph — single if-check addition. ConvoySignInScreen already built in Phase 1.
Risk	**MEDIUM — touches app entry point. Must not block offline operation.**
Est.	1 hour

5.2 Wire Ride Creation to POST /rides

ITEM	DETAIL
Connection point	ConvoyViewModel.kt createRide() function — replace local JSON write with ConvoyApiClient.createRide() POST. On success, write API response to convoy_events/ as local JSON cache.
Fallback	If API call fails (offline), fall back to local-only ride creation. User sees a banner: 'Ride saved locally — will sync when online'. Radio write still works.
V2.4 compatibility	Rides created in V2.4 remain valid. eventId UUID is preserved as rides.ride_id. Existing .convoy files continue to work unchanged.
Risk	**MEDIUM — touches createRide() in ViewModel. Fallback path must be solid before wiring.**
Est.	1.5 hours

5.3 Wire Invite Generation and Share Sheet

ITEM	DETAIL
Connection point	Ride detail screen — add SEND INVITE button. Calls ConvoyApiClient.createInvite(rideId). Opens Android ACTION_SEND share sheet with invite URL.
URL format	http://grouptrack.org/invite/{token} — links back to deep link handler built in Phase 1 Task A-04.
Risk	**LOW-MEDIUM — new button on existing screen. Isolated API call.**
Est.	1 hour

5.4 Wire Automated Map Download on Invite Accept

ITEM	DETAIL
Connection point	ConvoyInviteHandler.kt (Phase 1 Task A-04) — after saving ride JSON, check if map_bounds_north/south/east/west are set. If yes, call ConvoyTileDownloader.downloadTiles() for the ride area.
Organizer setup	Ride creation screen gains the existing DOWNLOAD REGION draw tool in a new context. Drawn bounds are stored in rides table via POST /rides.
Risk	**LOW — uses existing tile downloader. New trigger only.**
Est.	1.5 hours

6. Full Task Sequence — Ordered Build Plan

#	Phase	ID	Task	Est.	Risk
1	**Backend**	**B-01**	Create RDS schema — run create_schema.sql. 4 tables: users, rides, invites, enrollments.	30 min	**NONE**
2	**Backend**	**B-02**	Verify PHP endpoints with curl — POST /users/register → confirm RDS row created.	20 min	**NONE**
3	**Android**	**A-01**	Add API_BASE_URL to ConvoyConfig.kt. One line.	5 min	**NONE**
4	**Android**	**A-02**	Create ConvoyApiClient.kt — OkHttp wrapper for all endpoints. New file only.	90 min	**NONE**
5	**Android**	**A-03**	Add Google Sign-In dependency to build.gradle.	10 min	**NONE**
6	**Android**	**A-04**	Build ConvoySignInScreen.kt — Google One Tap, POST /users/register, cache user_id. NOT wired to nav yet.	2.5 hrs	**NONE**
7	**Android**	**A-05**	Add invite deep link handler — AndroidManifest intent-filter + ConvoyInviteHandler.kt. GET /ride/{token} → save to convoy_events/.	90 min	**LOW**
8	**Android**	**A-06**	Lead Lock UI buttons — SET AS LEAD (NODE panel), I AM LEAD + RECALC LEAD (MY CART panel). Logic already committed.	90 min	**LOW**
9	**Android**	**A-07**	Build and install. Full regression test of all V2.4 features before Wednesday.	30 min	**LOW**
⚠️ WEDNESDAY GATE — All FT tests must pass before proceeding to Phase 2					
10	**Phase 2**	**W-01**	Wire sign-in check into app launch — check SharedPreferences for user_id, navigate to ConvoySignInScreen if missing.	1 hr	**MED**
11	**Phase 2**	**W-02**	Test sign-in end to end — sign in → POST /users/register → RDS row created → user_id cached → app proceeds to map.	30 min	**MED**
12	**Phase 2**	**W-03**	Wire ride creation to POST /rides — replace local-only save with API call + local cache. Add offline fallback.	90 min	**MED**
13	**Phase 2**	**W-04**	Wire invite generation — SEND INVITE button → POST /rides/{id}/invite → share sheet with invite URL.	1 hr	**LOW**
14	**Phase 2**	**W-05**	Test invite flow end to end — organizer sends invite → rider taps link → app opens → ride downloaded → radio write succeeds.	1 hr	**MED**
15	**Phase 2**	**W-06**	Wire automated map download on invite accept — check map_bounds in ride JSON → trigger ConvoyTileDownloader.	90 min	**LOW**
16	**Phase 2**	**W-07**	Full V3.0 end-to-end test — sign in, create ride with map area, send invite, rider accepts, tiles auto-download, radio write, mesh operation.	1 hr	**MED**

7. Separation Rules — What Must Never Be Crossed

These rules enforce the isolation between Phase 1 and Phase 2. They protect V2.4 stability during the build period before Wednesday.

RULE	RATIONALE
ConvoySignInScreen must NOT be in the nav graph until Phase 2 Task W-01	An unreachable screen cannot break launch flow. The moment it is wired in, every app start depends on it.
ConvoyApiClient must NOT be called from any existing ViewModel until Phase 2	All existing ViewModel code operates offline. Introducing network calls before Wednesday risks timeouts and crashes during field test.
createRide() must NOT call the API until Phase 2 Task W-03	The existing ride creation flow is proven working. Do not touch it until Wednesday confirms V2.4 is solid.
build.gradle Google Sign-In dependency may be added in Phase 1 — it is a library, not a code path	Adding a dependency to build.gradle has zero runtime impact if nothing calls it. Safe to add any time.
AndroidManifest deep link intent-filter is safe in Phase 1 if handler gracefully no-ops when user_id is missing	A deep link that silently does nothing when prerequisites aren't met cannot break the existing app.
Radio write engine, ConvoyProfileBuilder, and InstallProfileUseCase must NOT be touched at all in V3.0	The radio write process is the most complex proven component. V3.0 only changes where ride JSON originates, not how it is applied.

8. Environment Reference

ITEM	VALUE
Branch	feature/convoy-event-ride
EC2 IP	34.224.89.217 — convoy-api-server
EC2 SSH	ssh -i ~/.ssh/convoy-api-key-2.pem ec2-user@34.224.89.217
RDS Endpoint	convoy-tracker-db.cudtjxrtdbql.us-east-1.rds.amazonaws.com
RDS Database	convoy_tracker / convoy_admin
API Base URL	http://34.224.89.217/
API File (EC2)	/var/www/html/convoy_api.php
Web Root	/var/www/html/
Website	http://grouptrack.org — LIVE
Device 1	8624SBCEDF00001789 — primary test device
Device 2	24039703201775
Key pair	~/.ssh/convoy-api-key-2.pem
Google Play Services dep	com.google.android.gms:play-services-auth:21.x.x